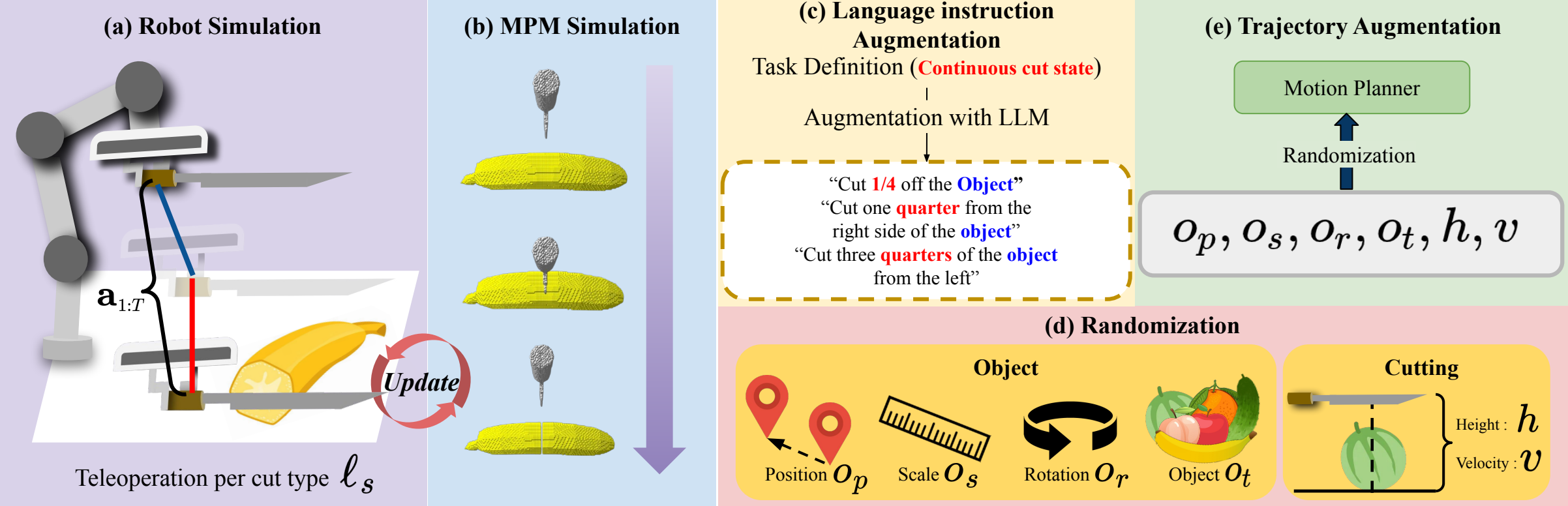
(a) Robot Simulation
$\mathbf{a}_{1:T}$
Update
Teleoperation per cut type ℓ_s
(b) MPM Simulation
(c) Language instruction Augmentation
Task Definition (Continuous cut state)
Augmentation with LLM
"Cut 1/4 off the Object"
"Cut one quarter from the right side of the object"
"Cut three quarters of the object from the left"
(e) Trajectory Augmentation
Motion Planner
Randomization
o_p, o_s, o_r, o_t, h, v
(d) Randomization
Object
Position o_p
Scale o_s
Rotation o_r
Object o_t
Cutting
Height : h
Velocity : v